

TPL 8-32

Serial No. BDM0000050–



Thank you for choosing Väderstad as your supplier!
We hope that our products will increase your profitability
and contribute to you having successful crops on your farm.

Warm regards,
The Stark Family

Contents

1	Declaration of conformity and machine identity	1	11.3	TPL 12	23
1.1	Declaration of conformity.....	1	11.4	TPL 16	24
1.2	UK Declaration of conformity.....	2	11.5	TPL 18	25
1.3	Nameplate.....	3	11.6	TPL 24	26
1.4	Technical data	4	11.7	TPL 32	27
2	General safety regulations.....	5			
2.1	Obligations and responsibilities	5			
2.2	Before using the machine	5			
2.3	How to read these instructions.....	5			
2.4	Description of Safety Symbols	5			
2.5	Safety Instruction.....	6			
2.6	Scrapping and recycling.....	7			
3	Installation	8			
4	General settings.....	9			
4.1	Output settings	9			
4.2	Change roller in the BioDrill 500	10			
4.3	Installation of ISOBUS/E-Control in the tractor.....	10			
5	Control system.....	11			
6	Filling and emptying.....	12			
6.1	Before filling the hopper	12			
6.2	Fill the hopper.....	12			
6.3	Opening the discharge doors	13			
7	Calibration.....	14			
7.1	Calibrating the BioDrill 500.....	14			
8	Maintenance and service	15			
8.1	Regular maintenance	15			
8.2	Seed transport	16			
8.3	Seed meter	16			
8.4	Clean the feed system	18			
8.5	When in storage for long periods.....	18			
9	Troubleshooting.....	19			
9.1	Alarm list.....	19			
10	Seed table	20			
11	Location of seed hoses.....	21			
11.1	TPL 8.....	21			
11.2	TPL 10	22			

1 Declaration of conformity and machine identity

1.1 Declaration of conformity



EC declaration of conformity

Väderstad AB, PO Box 85, SE-590 21 Väderstad, SWEDEN

hereby declares that the products named below have been manufactured in conformance with the Council's directive

2006/42/EC

and

2014/30/EU

The declaration above covers the following machines:

BioDrill 500

serial no: BDM0000050–BDM0005000

Väderstad 01-07-2023

A handwritten signature in black ink, appearing to read 'Gunnar Blackert'. The signature is fluid and cursive, with the first name 'Gunnar' and the last name 'Blackert' clearly distinguishable.

Gunnar Blackert

Technical Product Manager Drills

Väderstad AB

PO Box 85, 590 21 Väderstad

The undersigned is authorised to provide technical documentation for the machines named above.

1.2 UK Declaration of conformity

Applicable for machines with UKCA marking



UK declaration of conformity

Vaderstad AB, PO Box 85, SE-590 21 Vaderstad, SWEDEN

hereby declares that the products named below have been manufactured in conformance with

The Supply of Machinery (Safety) Regulations 2008

and

Electromagnetic Compatibility Regulations 2016

The declaration above covers the following machine:

BioDrill 500

Serial no: BDM0000050–BDM0005000

Vaderstad 01-07-2023

A handwritten signature in black ink, which appears to read 'Gunnar Blackert'.

Gunnar Blackert

Technical Product Manager Drills

Vaderstad AB

The undersigned is authorised to provide technical documentation for the machines named above.

1.3 Nameplate

VÄDERSTAD

Type Model year Serial No./VIN

Designation Working width Transport width

Basic weight Max. total weight Max. payload

Max. axle load Max. coupling load Mfg. year

498789
Väderstad AB, Box 85, SE-590 21 Väderstad

CE

Figure 1.1

- A. Machine type
- B. Serial number (always quote your serial number when ordering spare parts or when you are having your machine serviced or making a complaint)
- C. Year of manufacture
- D. Working width
- E. Transport width
- F. Tare weight of the basic machine
- G. Maximum total weight
- H. Maximum permitted payload
- I. Maximum permitted axle load
- J. Maximum coupling load (at the tractor's hitching point)
- K. Model year
- L. Use

Declaration of conformity and machine identity

1.4 Technical data

BioDrill 500	
Machine weight (kg)	260
Noise level, fan	Sound pressure, max 75 dB(A)

2 General safety regulations

2.1 Obligations and responsibilities

These instructions are to be regarded as for guidance only and entail no responsibility whatsoever on the part of Väderstad AB and/or its representatives. Full responsibility for the use, transportation, maintenance and servicing of the machine lies with the owner/driver. These instructions cover a large number of machine configurations.

Local conditions affecting crop rotation, soil type, climate, etc. may require procedures different to those mentioned in these instructions.

It is the full responsibility of the owner/driver to ensure that the machine is used correctly in all respects. The owner is also responsible for ensuring that all individuals who use the machine have read and understood this instruction manual and that they work in accordance with all valid provisions and regulations.

If an individual working with the machine detects any safety defects, these must be addressed immediately.

All Väderstad machines have passed quality assurance inspections and operational tests prior to delivery. However, the user/owner shall bear full responsibility for the machine functioning correctly when in use in the field. In the event of a complaint, please refer to the "General delivery provisions for the Väderstad Group".

Design modifications form a part of the continual improvement of our machines. Descriptions of the machine therefore refer to the form and design of the machine at the time of writing. The user manual may contain images which show a machine that does not look exactly like the one you have received, depending on which optional additions have been fitted, the model, or whether any updates have been carried out. Some descriptions in these instructions may not be relevant to your machine.

The operating instructions are designed according to the guidelines of the ISO 3600 standard.

2.2 Before using the machine

- A. Carefully read through these instructions and ensure that you have understood their content.
- B. Learn to use the machine correctly and carefully! In the wrong hands or if carelessly operated, the machine can become a dangerous implement.
- C. The machine will form a part of your workplace and the workplace of your colleagues. It is therefore important to ensure that everyone is protected and that functioning safeguards are in place.

2.3 How to read these instructions

Letters within brackets refer to corresponding letters in the image and are used as a reference in a text.

A. Reference (A)

B. Reference (B)

Information for which the order is important is indicated using numbered action instructions.

When referring to images, figures are also used in the same way as the alphabetical list if the references outnumber the letters in the alphabet.

1. Start by...
2. Then ...

2.4 Description of Safety Symbols



Always pay extra attention to texts or figures indicated by this symbol. The symbol indicates dangers that **may result in** death, serious bodily harm or considerable material damage if actions are not taken to avoid them



Always pay extra attention to texts or figures indicated by this symbol. The symbol indicates dangers that **may result in** death, serious bodily harm or considerable material damage if actions are not taken to avoid them



A signal word used to indicate a potentially hazardous situation which, if not avoided, could result in minor or moderate injury. However, such distinctions can be too subtle to affect behaviour (or to be translated). In certain circumstances, signal phrases such as "DANGER OF DEATH", "RISK OF BLINDING", or "BEWARE OF FUMES" can be more effective at drawing greater attention to some instructions or safety information than signal words.



This indicates a special situation or an activity required for proper machine handling. Failure to follow these instructions may cause disruption to the machine or to its surroundings.



Information marked with this symbol may be worth noting as it relates to a useful tip or a particularly useful piece of information that will help you handle the machine properly.



Used to clarify information.

General safety regulations

- Used to present information in the form of bullet points. The order in which the information is set out is not indicative of any particular order of precedence which must be followed.

2.5 Safety Instruction



Make sure that the machine is set to the correct height. Correct height of the machine will ensure that the loading height is correct. The loading height varies according to the machine type.



Risk of eye injury! Never look into the radar optics while in operation!



Always follow the instructions and safety precautions for the host machine.



Always make sure that the couplings are clean and free from dirt.



To maintain the machine's high level of quality and operational safety, use only genuine Väderstad spare parts. The warranty and any claims under will become null and void if non-genuine parts are used.

2.6 Scrapping and recycling

When the machine or any of its parts are worn out, Väderstad AB recommends that scrapping be carried out according to applicable regulations in order to avoid toxic and environmentally hazardous materials ending up in the environment.

During its lifespan:

- Replaced wear parts in steel should be sent to metal scrapping companies.
- Replaced wear parts in rubber, plastic or the like should be sent to the appropriate recycling facility.
- Oil, oil filters, batteries, and other electrical components may have a negative impact on the environment and should be sent to the appropriate recycling facilities.

In the case of scrapping:

- The machine mostly consists of steel that is suitably scrapped by an authorised metal scrapping company.
- Other components should be taken care of according to local recycling regulations.

REACH Art. 33

- REACH is a regulation adopted to improve the protection of human health and the environment from the risks that can be posed by chemicals. For more information on how Väderstad AB follows REACH Article 33. Visit www.vaderstad.com/reach

Always follow current local recycling regulations and applicable ordinances.

3 Installation



For ISOBUS/E-Control, please refer to the separate user guide.

4 General settings

Applicable from serial number BDM0000050

4.1 Output settings

To adapt the feed to different types, grades, or amounts of seed, there are different rollers to choose from. See table “Table 4.1 Output settings for BioDrill 500 when installed on Tempo L”.



If the material used in the BioDrill 500 is damp, there is a risk of blockage in the hoses.

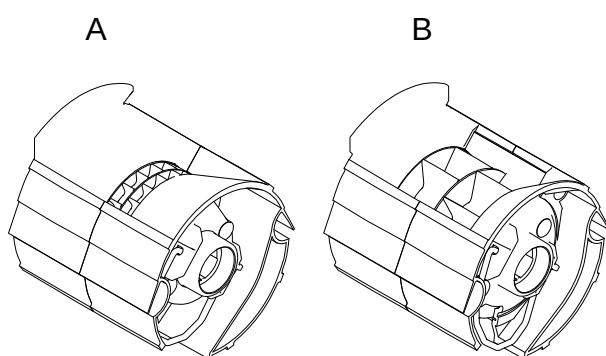


Figure 4.1

- Roller type A: Application rate 3–8 (kg/ha)
- Roller type B: Application rate 9–50 (kg/ha)

Table 4.1 Output settings for BioDrill 500 when installed on Tempo L

Roller	Application rate (kg/ha)	Fan rotation speed (rpm)	Fan rotation speed (rpm)	Fan rotation speed (rpm)	Fan rotation speed (rpm)
		12 row units	16 row units	18 row units	24 row units
A	3–5	2700	3100	3100	3100
A, B	6–15	3100	3100	3100	3100
B	16–35	3400	3400	3200	3200
B	36–50	4900	4000	3800	4300

4.2 Change roller in the BioDrill 500

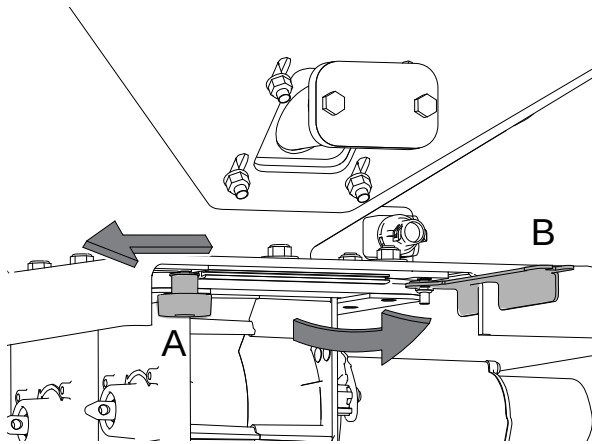


Figure 4.2

1. Turn the dial (A) and push to the left. Open the hatch (B) towards the side.

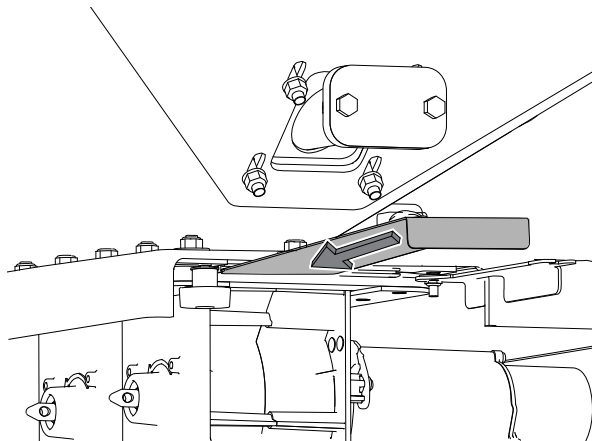


Figure 4.3

2. Insert the cover plate.

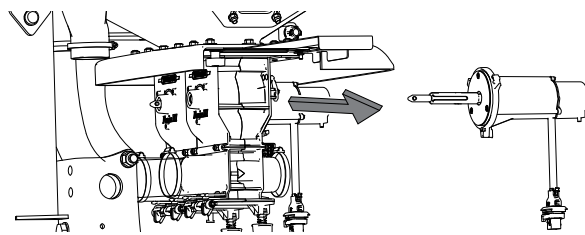


Figure 4.4

3. Turn and pull out the motor.
4. Change the roller and replace in the reverse order.

4.3 Installation of ISOBUS/E-Control in the tractor



For ISOBUS/E-Control, please refer to the separate user guide.

5 Control system



For ISOBUS/E-Control, please refer to the separate user guide.

6 Filling and emptying

6.1 Before filling the hopper

Check before filling

- That the machine is empty, clean and dry.
- That the discharge hatch is closed.
- That the correct rotor is fitted.
- That the level guard is at the correct height.

6.2 Fill the hopper



Safety first! Never walk or stand under a suspended load!



Make sure nobody is on the machine or in the hopper during delivery of the product to the machine. Do not approach the hopper unless absolutely necessary.



Avoid contact with and inhalation of seed dressing.

1. To fill the hopper use a loader with a pallet.

Place the bags on the pallet.

2. Set the machine to the correct loading height.

When at the correct loading height the ladder will be at a good angle and the correct distance from the ground so that the operator can easily access the platform.

3. Use the provided access way when approaching the platform.

6.2.1 Adjust the level guard

Location of the level guard

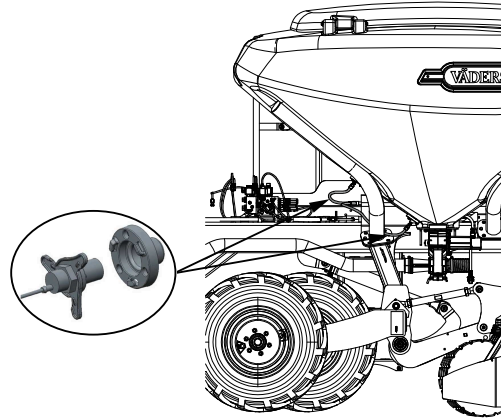


Figure 6.1

The height of the level guard can be adjusted so that the sensor can be moved between the different brackets.

- When drilling e.g. higher rates or a large size product, the level guard can be placed higher up in the hopper.
- When drilling low rates or a small size product, the level guard can be placed at a lower level on the hopper.

Disassembly and assembly

1. Turn the level guard counter clockwise and loosen from the bracket.
2. Turn the level guard clockwise to assemble on to the other bracket.

6.2.2 Level scale for seed

A scale with intervals of 50L is installed inside the hopper to measure the level of the product. The scale has a limit of 500L.

6.3 Opening the discharge doors

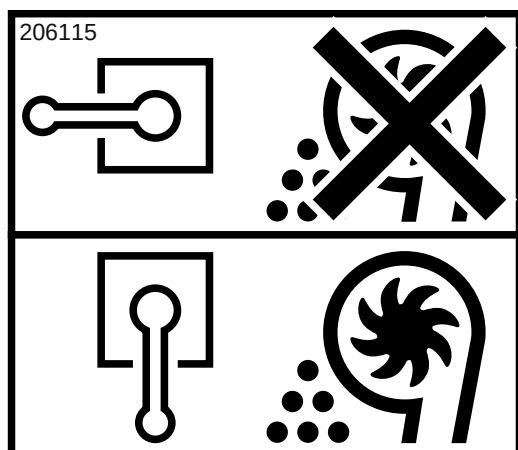


Figure 6.2



Before opening the discharge door check that the fan is turned off.

The seed feed housing is emptied in a single operation.

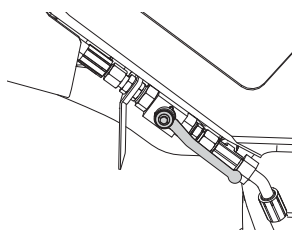


Figure 6.3 Fan lever

1. Turn off the fan by switching off the hydraulic remote on the tractor. If the agitator is required for efficient emptying, use the ball valve instead.

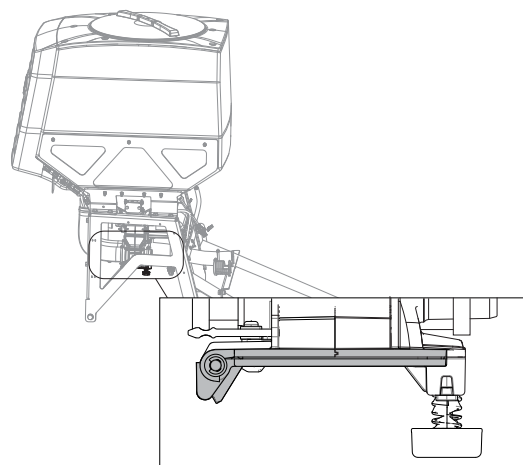


Figure 6.4

2. Turn the knob to release both the discharge doors.

7 Calibration



For ISOBUS/E-Control, please refer to the separate user guide.

7.1 Calibrating the BioDrill 500

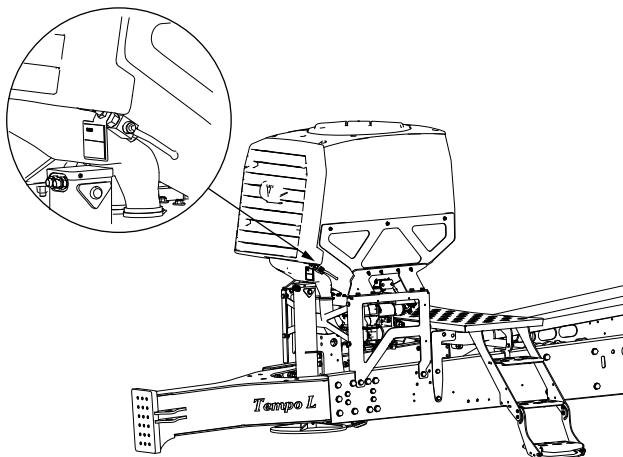


Figure 7.1

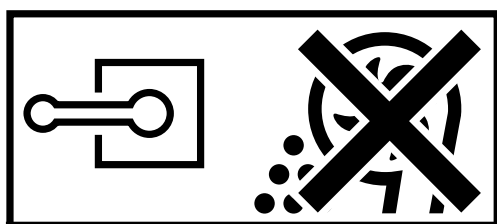


Figure 7.2

1. Turn the ball valve on the BioDrill 500 hopper to switch off the fan.
2. Start the hydraulic system for the BioDrill 500 hopper to rotate the agitator.

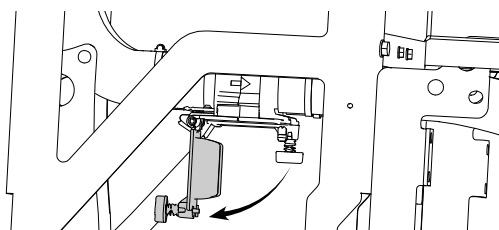


Figure 7.3

3. Open the latch by pushing and turning the grey knob, see "Figure 7.3"
4. To calibrate, follow the procedure for fertiliser calibration see separate manual
5. Close the hatch and shut off the hydraulic system for the BioDrill 500 hopper.

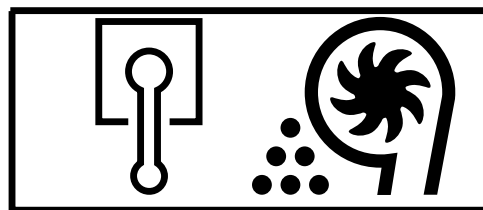


Figure 7.4

6. Switch on the fan with the ball valve.

8 Maintenance and service



Always shut down the tractor and remove the ignition key in conjunction with work using the hopper and when performing service and maintenance.



The machine must be supported safely on stands for all work underneath the machine or when there is a risk of crush injuries. Ensure also that the surface supporting the trestles is firm enough to support them.



Always lower the machine to the ground prior to performing any service on the hydraulic system!



Depressurise the hydraulic system before service and maintenance.



The machine should be in the lowered position and standing on a flat surface, when carrying out service and maintenance.



Always switch off the iPad (E-Control) when performing maintenance and service work on the machine.

8.1 Regular maintenance



To preserve the high quality of the drill, always use genuine Väderstad spare parts.

Perform maintenance checks at regular intervals and at the end of the season.

- Check that there is no seed or any bits of packaging remaining in the BioDrill hopper, seed feed housing, rotor or feed system.
- Check that there has been no seed or any bits of packaging in the seed hoses or in the outlets of the distributor head.
- Check that no seed has run down into the ejector tubes and associated air hoses.
- Check that the seed hoses and couplings are not trapped or damaged.

8.1.1 Cleaning



Cleanliness is of vital importance when undertaking any kind of work with the machine's hydraulic system.

Wipe with clean paper or cloths. Place parts on a clean surface (not directly on the work bench) Rinse before assembly, for example with a degreasing agent.



Never rinse the bearings using a high-pressure water supply!

It is important to grease the bearings after washing in order to remove any water that has remained behind.



Never clean electrical/electronic components directly with a high-pressure water jet.

Clean electrical/electronic components by blowing them with air or by wiping them with a slightly dampened cloth.



Exercise extreme caution when washing with high-pressure water in the area around the seed feed housing.

After cleaning, air and dry all components carefully before reassembly.



Keep the fan running for a while after cleaning, so that the whole system is blown dry.

Any remaining seed that sprouts can block the air and seed hoses.

Seed can also attract small rodents, which can damage the seed drill.

- Clean BioDrill hopper, rotor, seed feed housing, seed hoses, the tramlining motors and outlets in the distributor head.
- Clean the seed feed housing. Open and disassemble according to "8.3 Seed meter". Allow all the parts to dry thoroughly prior to reassembly.
- Clean the radar.

8.2 Seed transport

8.2.1 Distributor head, FLEX

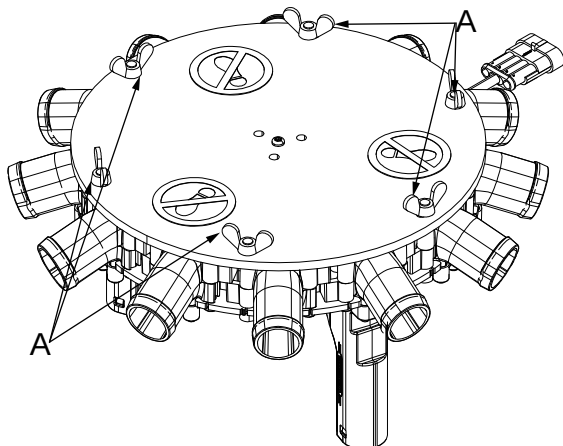


Figure 8.1

1. Open the cover by loosening and removing the six wing nuts (A).

Replace the distributor head's tramlining motors

Replace the distributor head's tramlining motors (B) if needed.

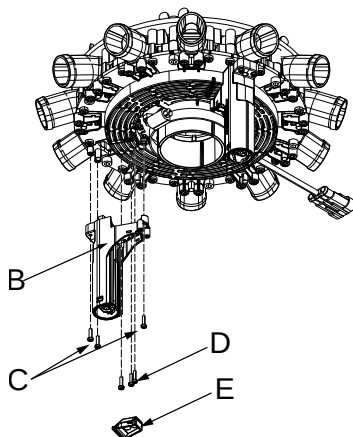


Figure 8.2

1. To remove a faulty motor from the distributor head, loosen and remove the three screws (C). Remove the rubber lid (E) and the three electrical screws on the circuit board (D).

Use the Torx screwdriver supplied.

2. Fit a new motor with the three electric screws (C) and the three electric screws (D). Use the Torx screwdriver supplied to tighten the electrical screws.
3. Assemble the rubber lid (E).

8.2.2 Repair and replace the seed hose

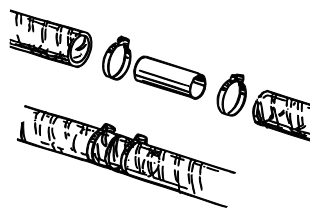


Figure 8.3

8.2.2.1 Repair

If a seed hose is damaged due to abrasion or folding, it can be jointed with a sleeve.

1. Cut the hose at the centre of the fold or damage.

Where necessary, the smallest possible part of the hose can be cut away. If the hose becomes too stiff at the joint and cannot be bent sufficiently when the machine is lowered, it may be necessary to replace the entire seed hose, or to replace part of the hose and joint it in two places.

8.2.2.2 Changing a seed hose

Use soapy water on the hoses when fitting them to seed coulters. Twist the hose anti clockwise when removing or fitting. This will help the spiral reinforcement to "open" the hose a little. Cut a replacement hose to the same length as the piece it is to replace.

8.3 Seed meter

BioDrill 500 is fitted with an electrically powered Fenix III seed feed housing. Monitoring and manoeuvring can be done using E-Control.

8.3.1 Settings

Sliding hatch

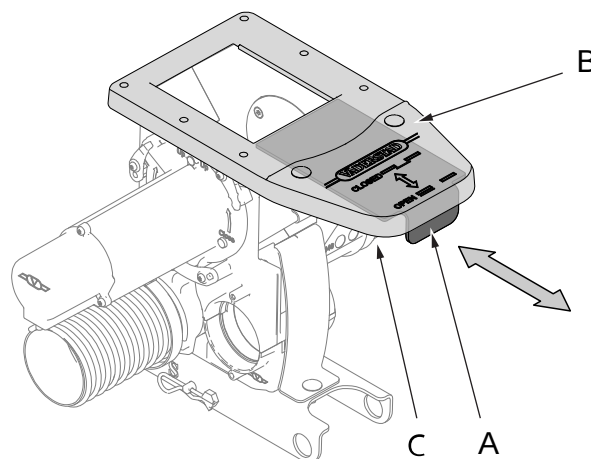


Figure 8.4

When the sliding hatch (A), located beneath the sealing cover (B), is pressed in, the supply from the hopper to the metering unit is blocked.

When the supply from the hopper is blocked, it is possible to replace the roller or perform service on the seed feed housing, even though the hopper is full.

During drilling, the sliding hatch must always be completely open and the wing nut (C) tightened.

8.3.2 Disassembly and assembly

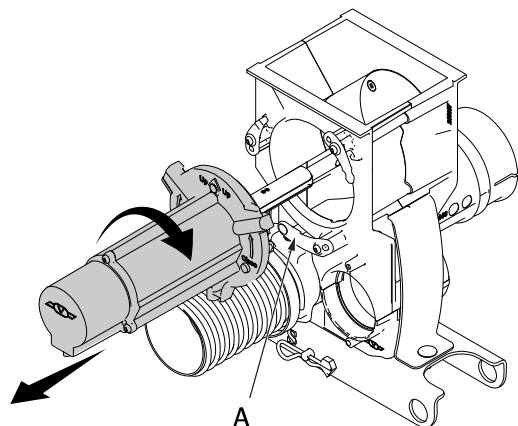


Figure 8.5

1. Open the catch (A) and turn the motor in a clockwise direction.
2. Remove the motor from the seed feed housing.

Changing a feed roller

Before changing the roller, it may be necessary to empty the lower part of the seed feed housing to gather up the seed.

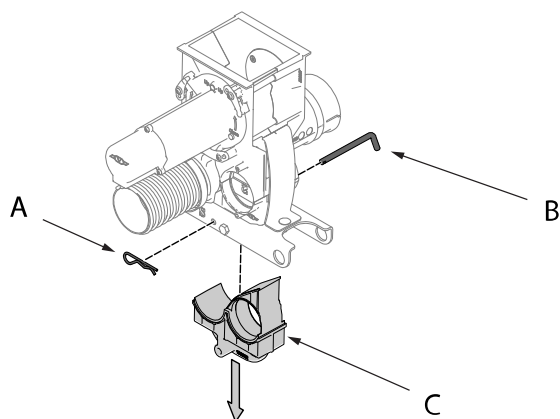


Figure 8.6

1. Loosen the locking pin (B) and pull out the cotter pin (B).
2. Pull the plug out.

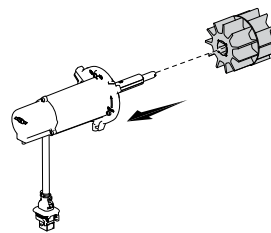


Figure 8.7

3. Separate the roller and the motor.
4. Switch to the required roller.

Note that there are two grooves along the motor shaft that must match the roller.

Note that the roller can only be attached in one direction.

Fit the output motor



After cleaning, carefully air and dry all components in the seed hopper before reassembly

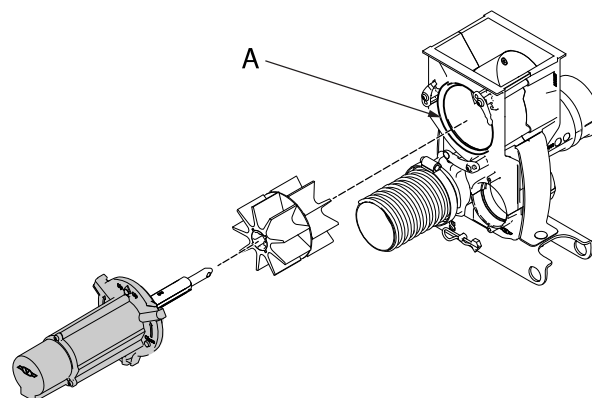


Figure 8.8

1. Check that the seed feed housing is empty. Clean inside when necessary. Clean outside the flange (A) on the outside of the housing.

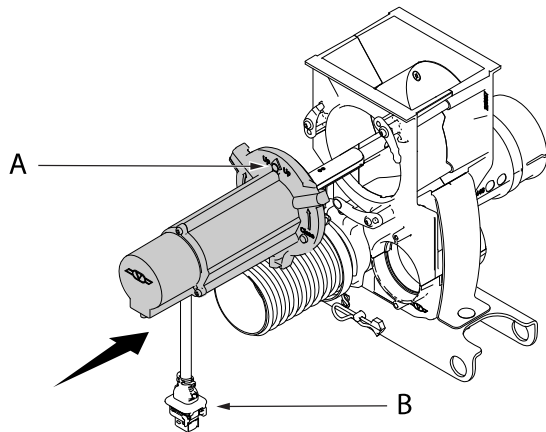


Figure 8.9

2. Insert the motor and roller in the seed feed housing.
The motor can only be fitted with the motor cable (B) downwards and the arrow (A) upwards.
3. Turn the motor counter-clockwise until it clicks into place.

Remove the plug in the seed feed housing

1. Loosen the locking pin (A) and pull out the cotter pin (B) in "figure 8.9".
2. Pull out the plug (C).

Fit the plug in the seed feed housing

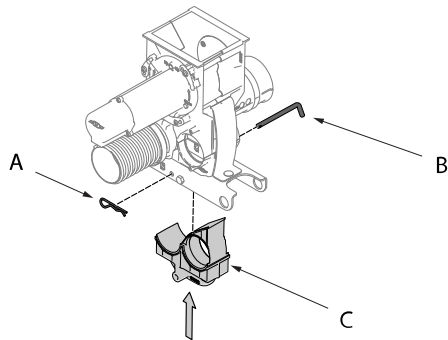


Figure 8.10

1. Attach the plug (C) to the seed feed housing.



Attach the plug in the correct direction.

Check that the pinhole in the plug is in line with the pinhole in the plate.

Check that the entire plug is inserted completely. The flat surface of the plug must be on the same level as that of the U-profile.

2. Assemble the cotter pin (B).
3. Assemble the locking pin (A).

8.4 Clean the feed system

The feed system on the drill should be cleaned regularly and always at the end of the season.

Seed residues attract small rodents.

8.5 When in storage for long periods

When not in use, the machine should be stored indoors to protect the machines electronic instruments. These electronic instruments are of a very high standard and have a high tolerance for moisture, we nonetheless recommend storage indoors.

Make sure that the seed drill has been emptied and thoroughly cleaned.

The feed rollers should be completely dry before being put into storage and should be kept in a dry, dark and rodent-free place when the machine is not in use.

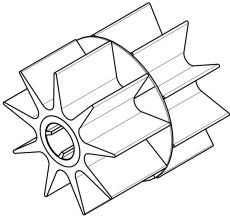
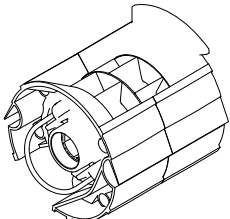
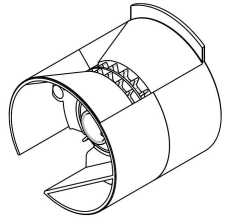
9 Troubleshooting

9.1 Alarm list



For ISOBUS/E-Control, please refer to the separate user guide.

10 Seed table

Crop	Driving speed km/h ¹	Seed delivery kg/ha ²	The seed output roller spare part numbers and volumes
Wheat	5–15 ³	70–400	 400 cm ²
Rye		70–350	
Corn		65–350	
Oats		55–280	
Peas	5–12 ⁴	60–325	
Beans		70–425	
Ley seed Flax	5–15	4–40 8–100	 120 cm ²
Ley seed Flax Rape seed		2–9	 18 cm ²

1. It may be possible to achieve higher driving speeds by limiting the interval for possible seed delivery.
2. Greater seed delivery may be possible by limiting the speed range.
3. Intervals for seed output (kg/ha) apply for the speed range 5–12 km/h for these crops.
4. Intervals for seed output (kg/ha) apply for the speed range 5–12 km/h for these crops.

11 Location of seed hoses

11.1 TPL 8

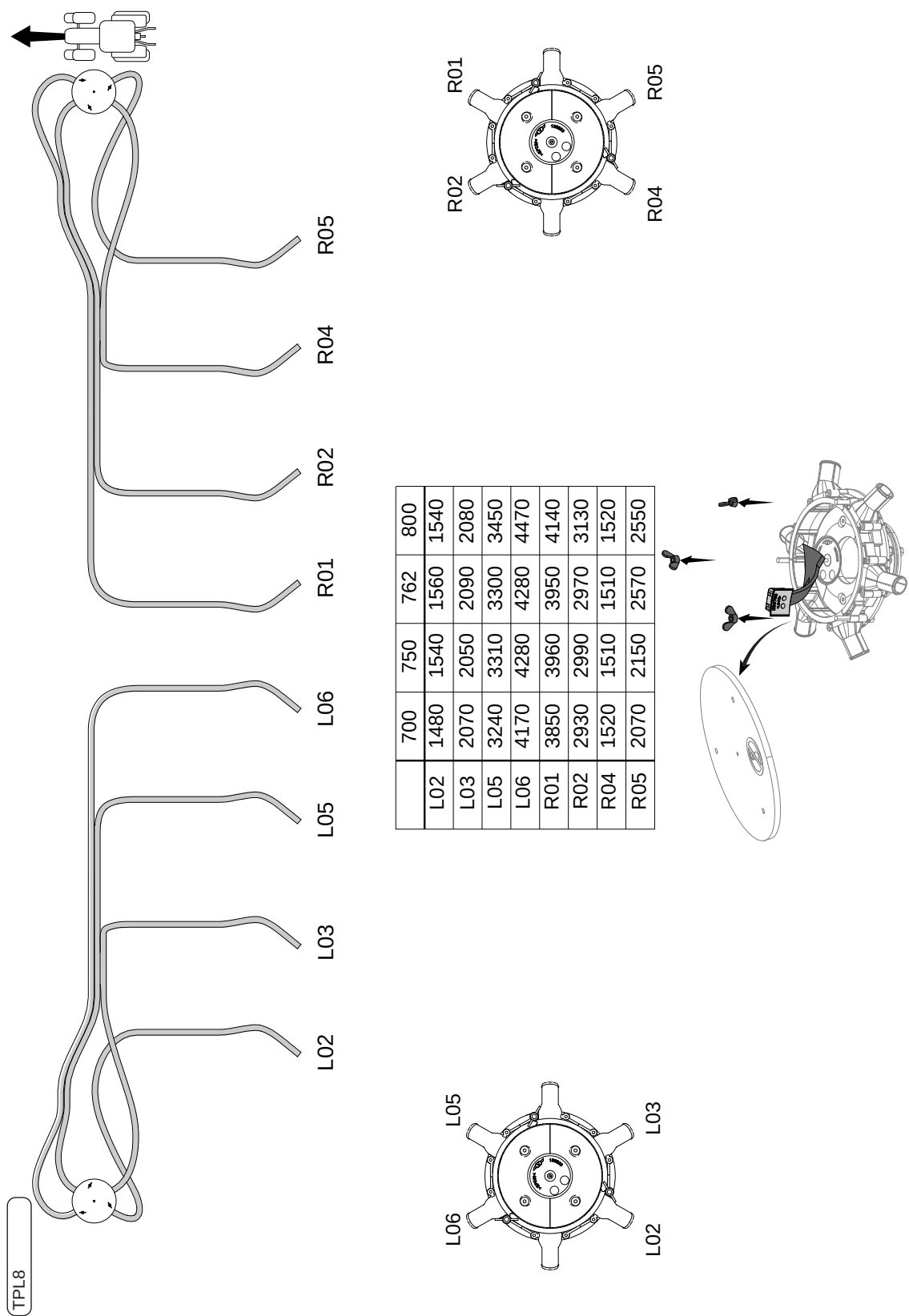


Figure 11.1

11.2 TPL 10

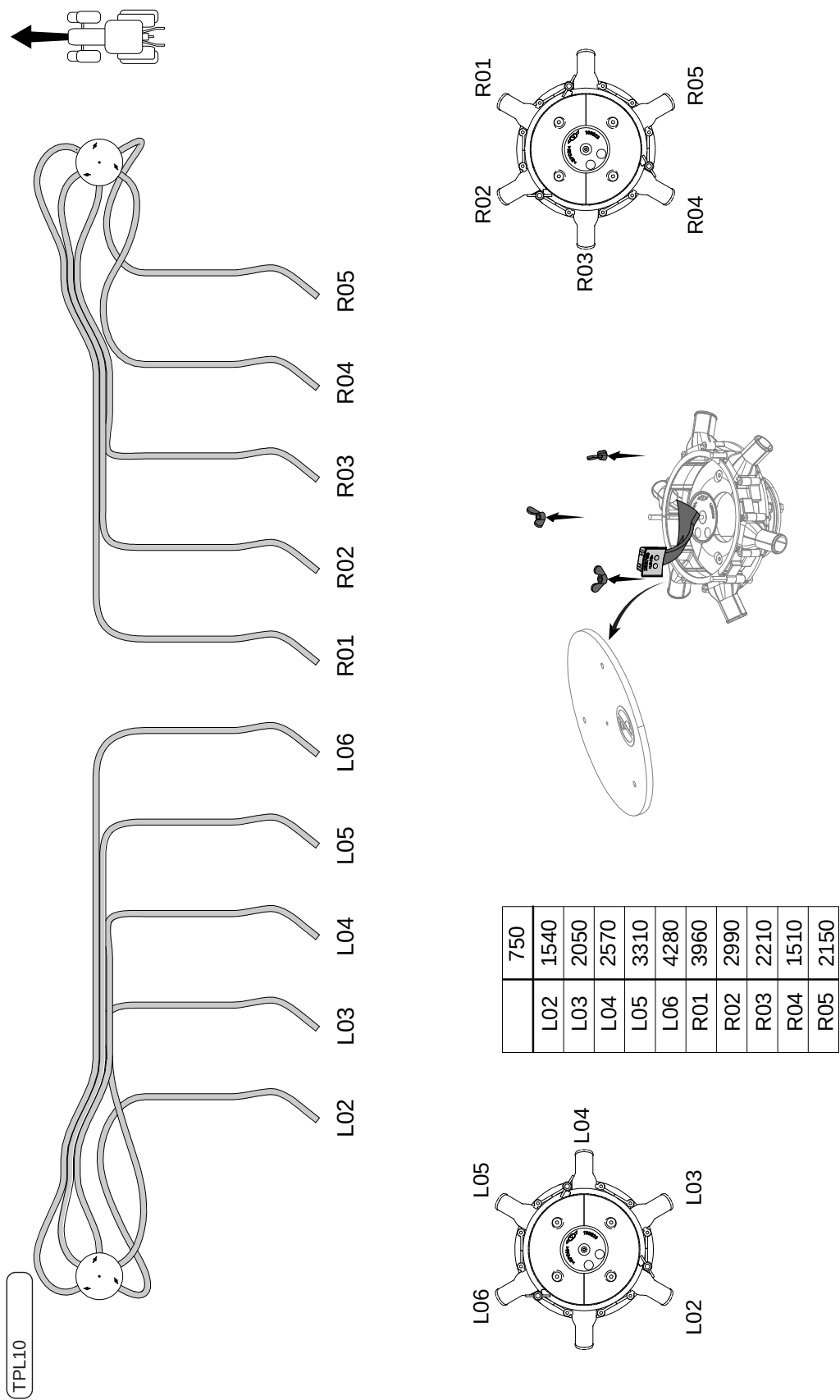


Figure 11.2

11.3 TPL 12

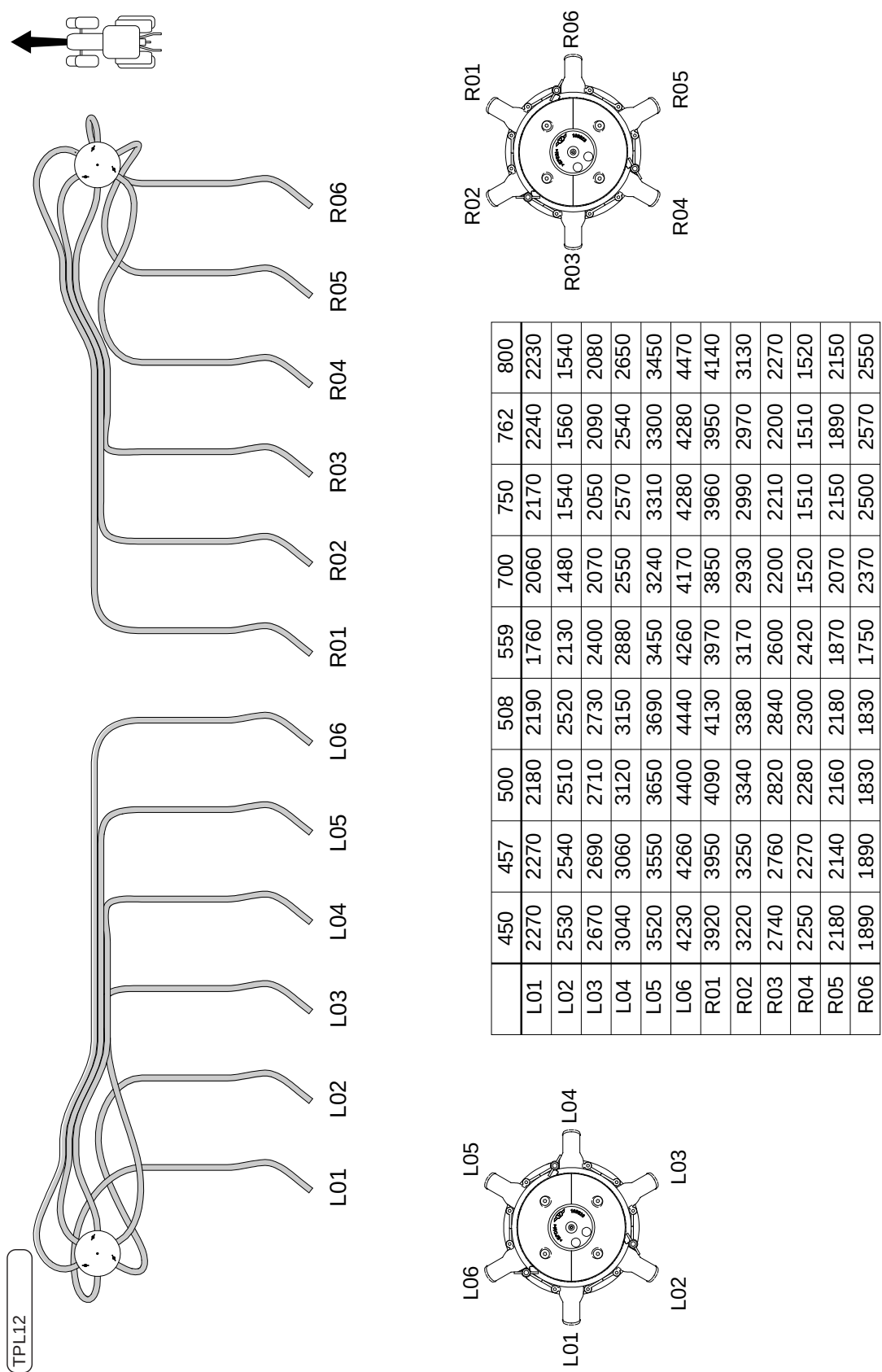


Figure 11.3

11.4 TPL 16

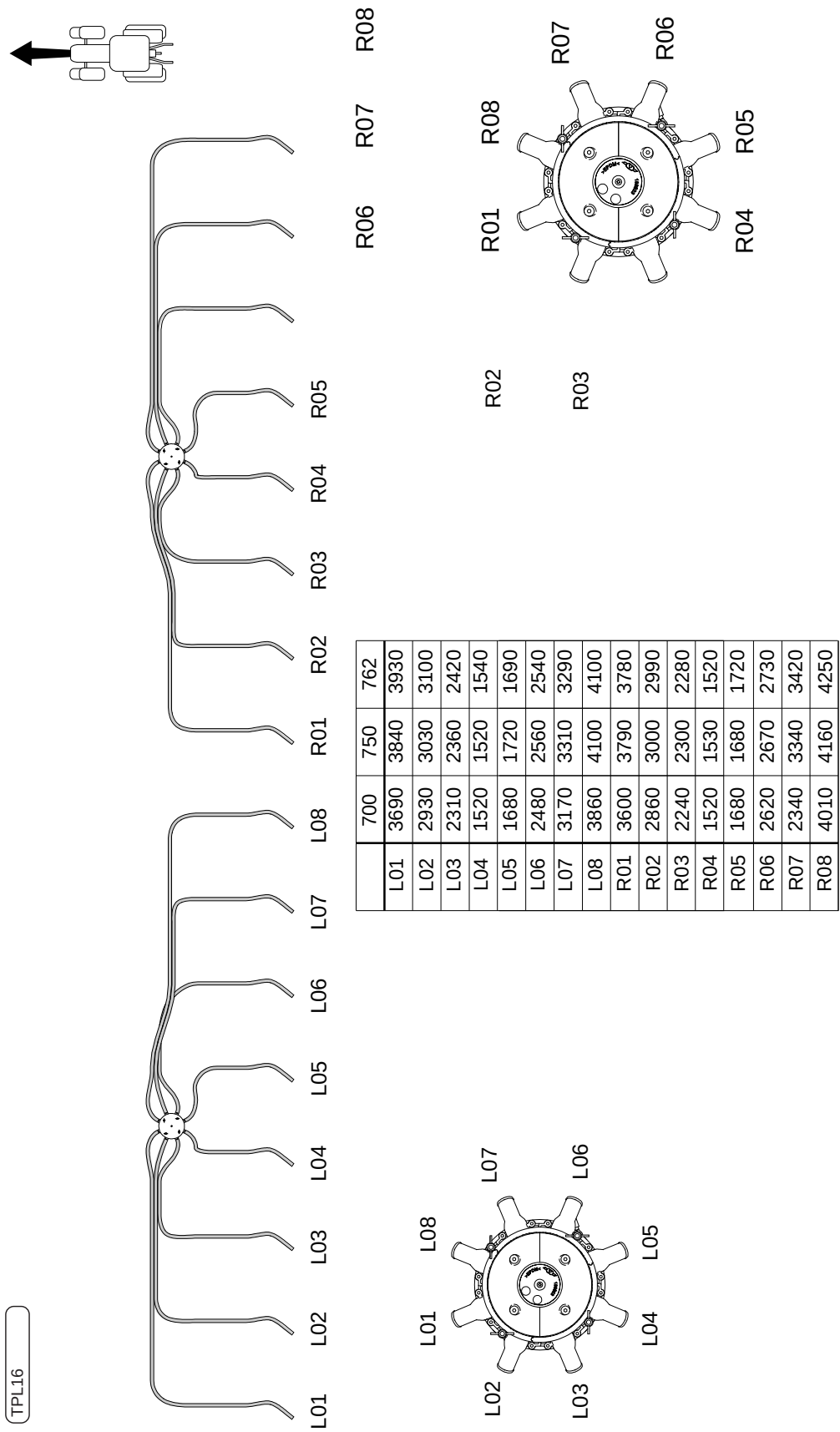


Figure 11.4

11.5 TPL 18

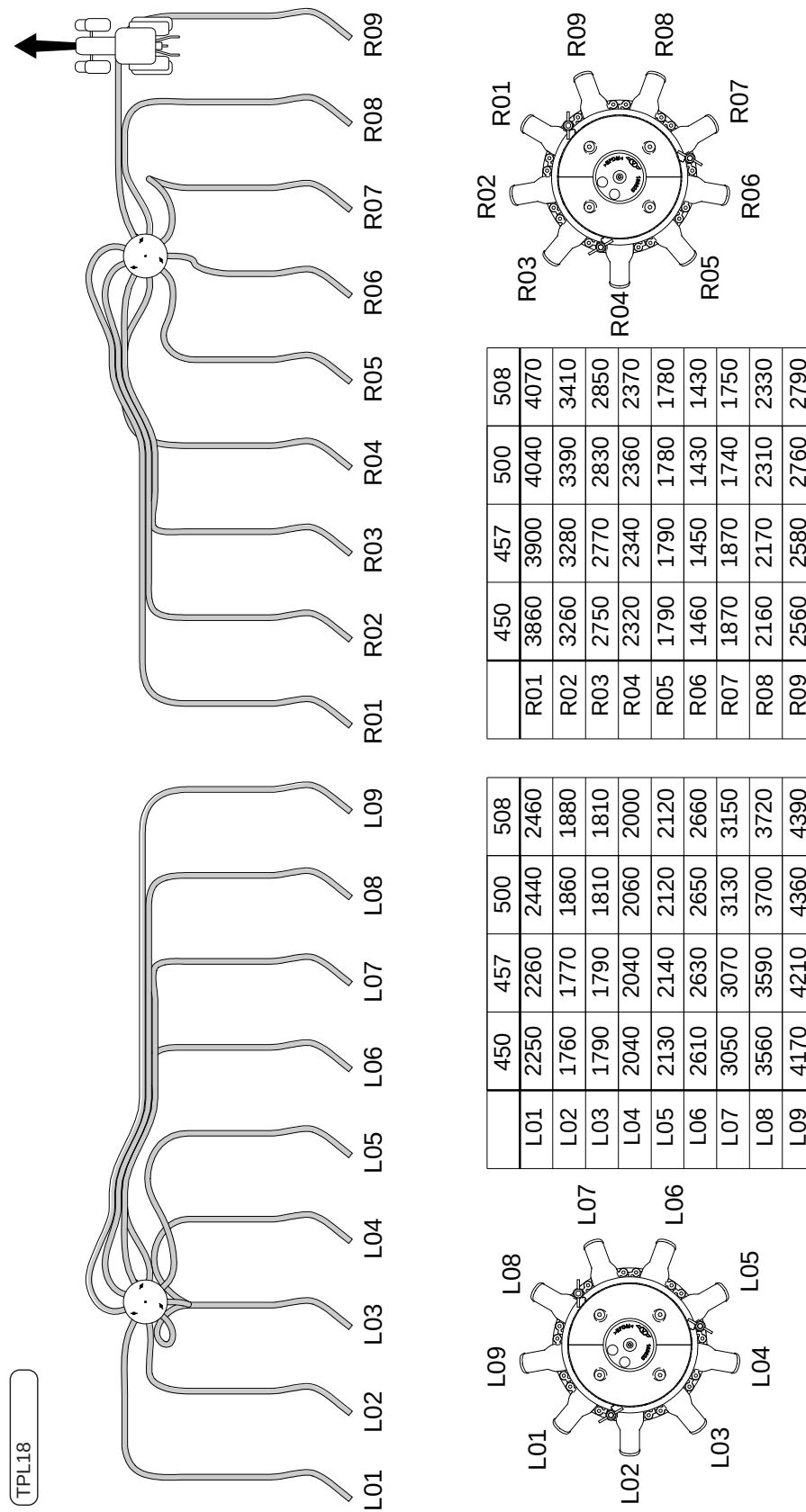


Figure 11.5

11.6 TPL 24

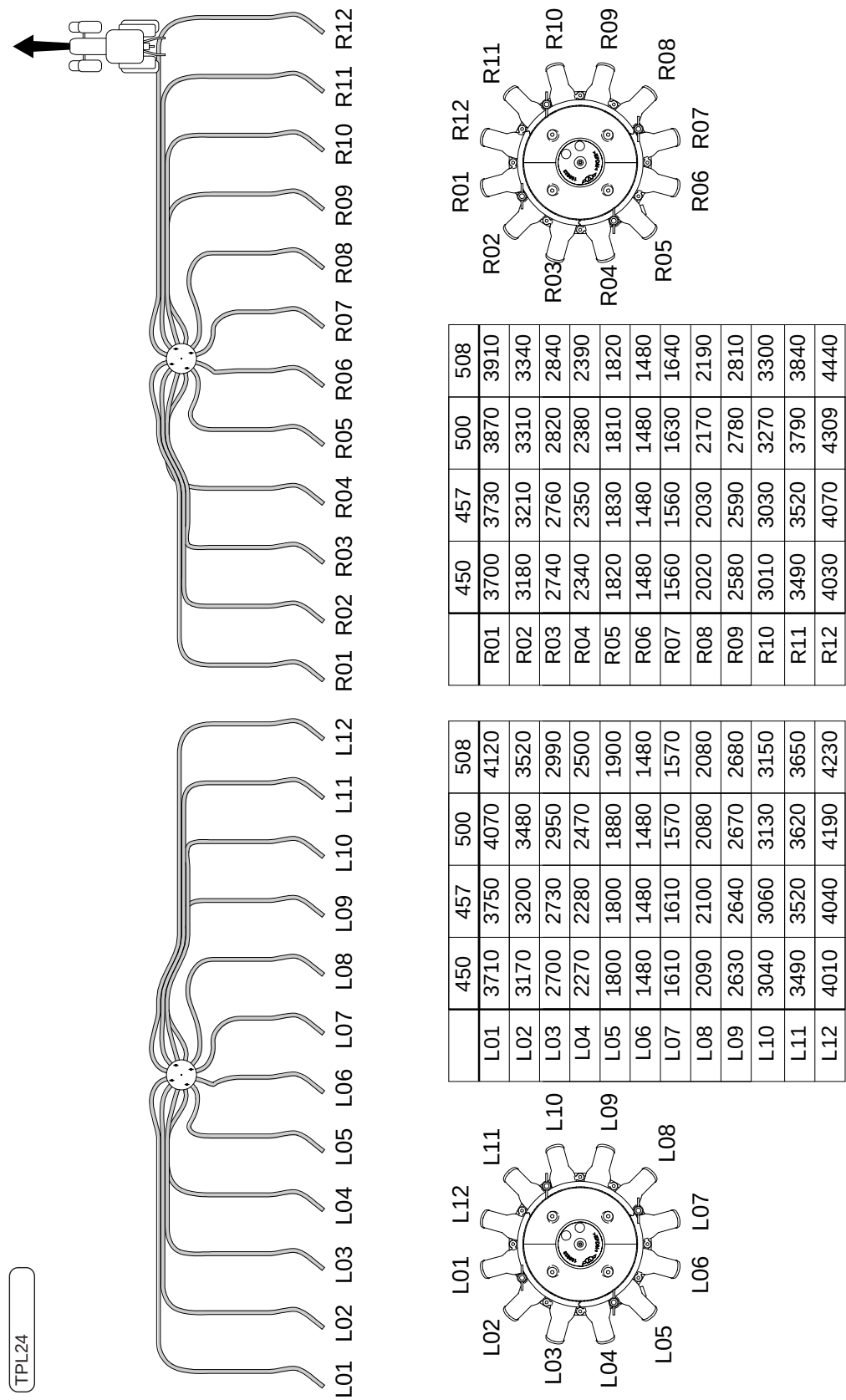


Figure 11.6

11.7 TPL 32

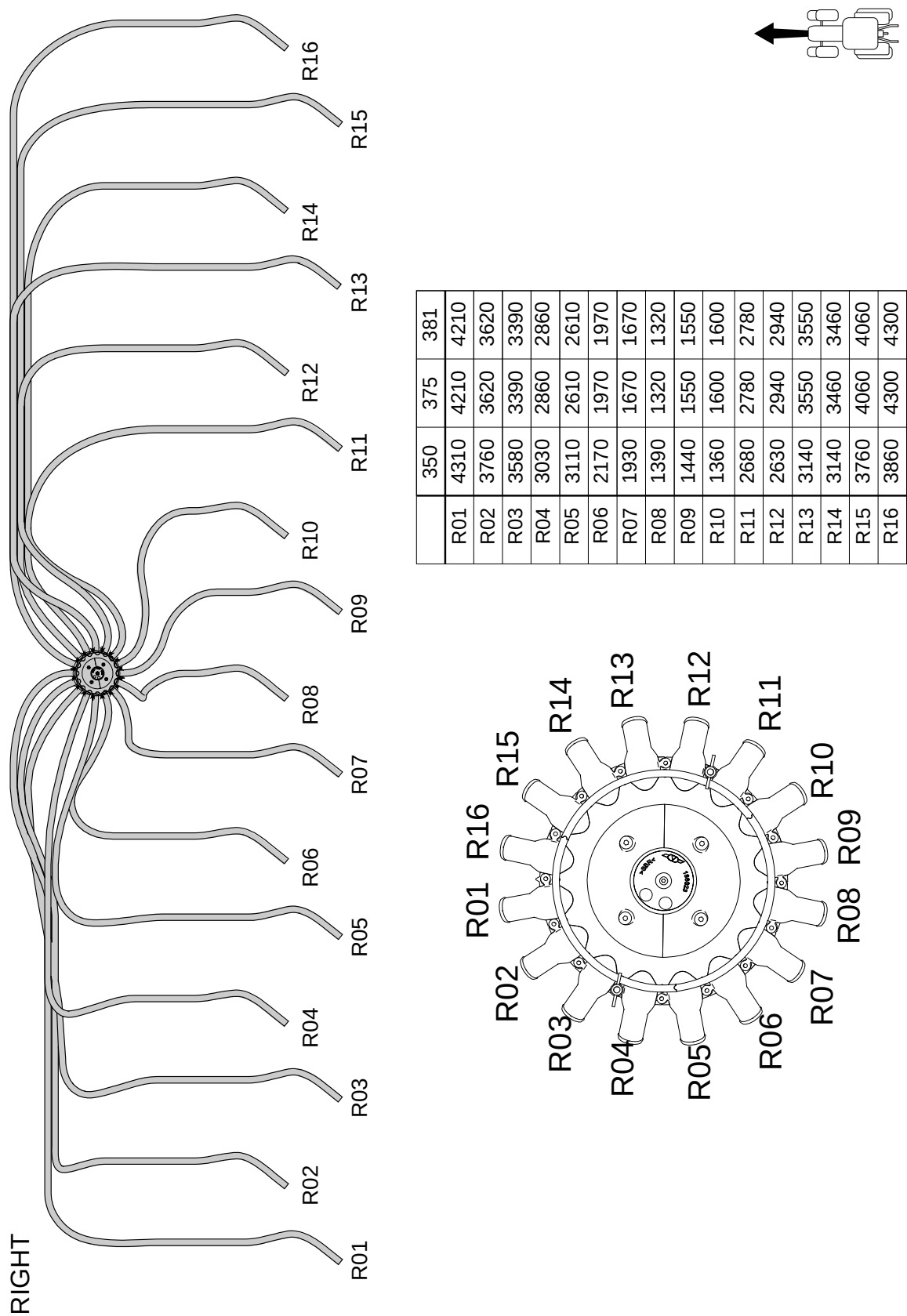


Figure 11.7

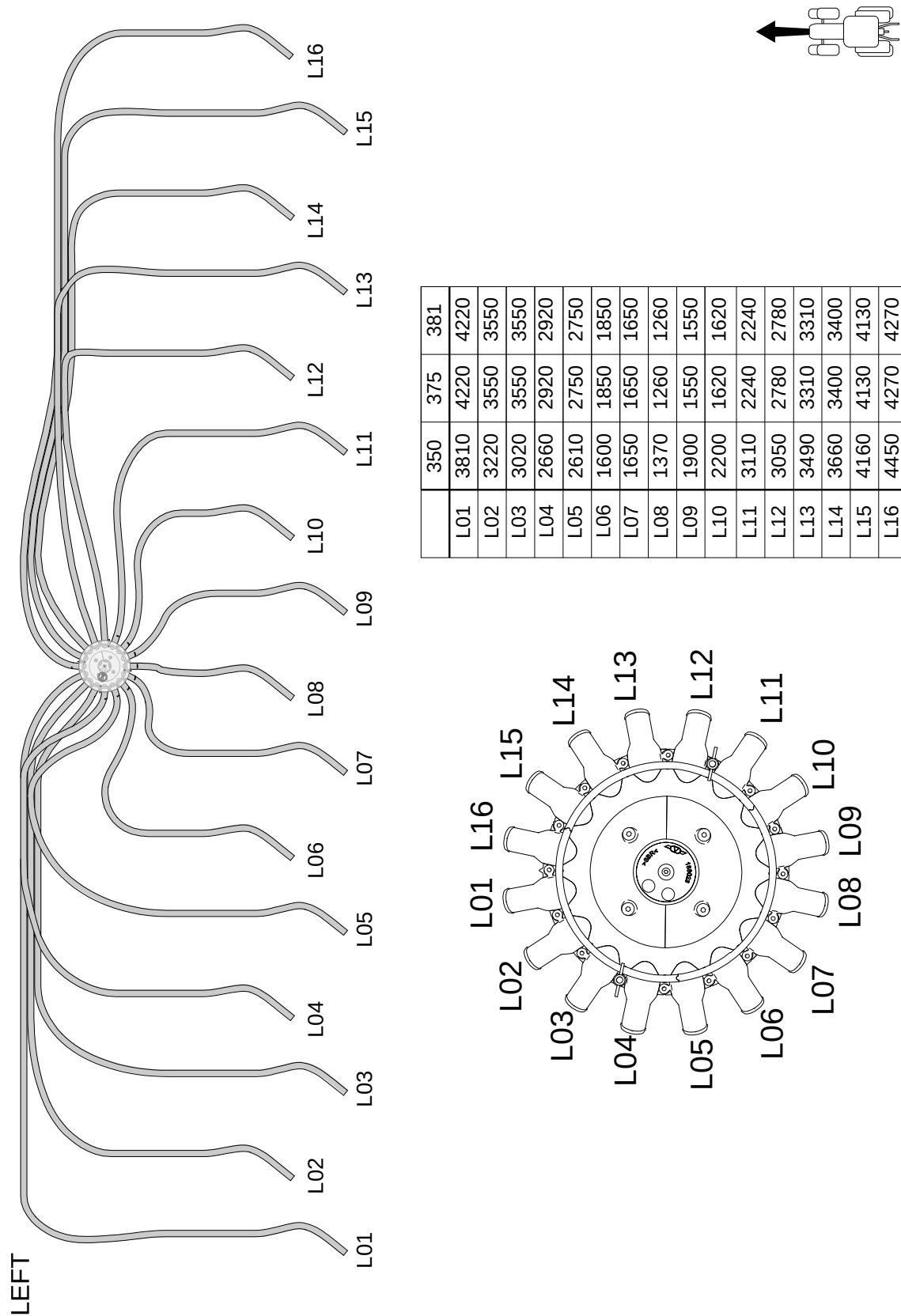


Figure 11.8

Väderstad AB
SE-590 21 VÄDERSTAD
Sweden
Phone: +46 142- 820 00
www.vaderstad.com

